Purabh Singh

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Education

Rutgers University Aug 2023 – May 2027

B.S. in Computer Science, Minor in Data Science *New Brunswick, NJ*

- **Relevant Coursework:** Data Structures, Algorithms, Computer Architecture, Discrete Structures I & II, Intro to AI, Systems Engineering, Data Management for Data Science
- **Achievements:** MusaLabs Hackathon Winner (1st of 50+ teams) · Dean's List

Experience

Software Engineering Intern June 2026 – Aug 2026

BNY *Pittsburgh, PA*

- Built an immutable fintech payments audit trail on a private Hyperledger Besu blockchain (Solidity, Web3j), validating and recording $2B+ in cleared debits and credits for real-time reconciliation across 3 fragmented systems with zero payment-latency impact, tested end-to-end using Apache ActiveMQ Artemis to simulate live message-queue traffic, fronted by an Angular/ethers.js UI that reads directly from Besu for live ledger visibility during backend downtime.
- Developed a real-time SWIFT MT/MX "blotter" dashboard for an Agile payments team, tracing every message as it moved through the platform's Java/Spring Boot systems, delivering first-ever end-to-end visibility into message flow across 15M+ daily messages.
- Built an Angular observability and troubleshooting dashboard for cross-functional ops teams, pulling from Splunk logs to monitor the health and security posture of all 11 payment-platform systems, surfacing per-system vulnerability counts and month-over-month trends and auto-generating Jira tickets from flagged issues to cut manual triage time by 60%.

Software Engineering Intern May 2025 – Aug 2025

Essex Securities *Jersey City, NJ*

- Built an AI agent using LangChain and RAG (pgvector semantic search, FastAPI endpoints) to automate extraction and comparison of actual vs estimated EPS from quarterly earnings reports, cutting manual review time by over 70%.
- Built an MVP BTC-USD spot exchange in Python (FastAPI, asyncio, PostgreSQL) with a price-time-priority matching engine supporting limit/market orders, partial fills, self-trade prevention, and sequenced WebSocket market data.
- Designed an append-only double-entry ledger with atomic per-trade settlement, pre-trade risk holds, and crash recovery that rebuilds the order book, with automated invariant checks proving every balance reconciles.
- Automated limit order execution via Bloomberg EMSX API integration, reducing manual entry time by 20%.

Software Engineering Intern May 2024 – Aug 2024

Azonix Tech Private Ltd *Remote*

- Built a real-time facial-recognition pipeline on a live 30 FPS edge video feed, using OpenCV and an ONNX-optimized InsightFace embedding model to detect and identify enrolled individuals in under 45 ms per frame at 96%+ accuracy.
- Engineered an IoT telemetry platform ingesting 5,000+ MQTT messages/sec from 500 concurrent sensor devices, persisting time-series data to TimescaleDB for sub-second querying and streaming anomaly detection.
- Trained an ML (XGBoost) predictive-maintenance model on 100K+ rows of device telemetry to forecast equipment failures roughly 6 hours in advance, reaching 0.92 AUC and cutting projected unplanned downtime by 40%.

Projects

LeaseLink June 2025

- Built a full-stack Tinder-style housing marketplace connecting renters and landlords through mutual interest, with a React + Tailwind frontend, Express.js backend, PostgreSQL database, and Google OAuth authentication.
- Led backend development of a gated real-time chat system via WebSockets and implemented landlord approval workflows, streamlining property discovery and communication.

FastRust Feb 2026 – Present

- Created an open source, from-scratch Rust web framework on Hyper and Tokio that pairs FastAPI-style ergonomics with Rust-native performance, featuring a custom trie data structure for routing with static-over-parameter precedence, typed extractors (Path, Query, JSON, State), and macro-based routing across 7 HTTP methods.
- Layered on an onion-style middleware stack (CORS, bearer auth, logging) with OpenAPI 3.1 generation and a live Swagger UI, benchmarked with Criterion at sub-microsecond trie-router lookups (172 ns static, 849 ns parameterized) and 1.4 to 2.3 μs request dispatch.

Technologies

Languages: Python, Java, TypeScript, SQL, C++, C, Rust, JavaScript, HTML | **Frontend**: React, Angular, Swift, Next.js, Tauri | **Backend**: Node.js, NestJS, LangChain, Express, FastAPI, Spring Boot, Pandas, Kafka, ActiveMQ Artemis, Jira | **DevOps/Cloud**: AWS, Azure, Docker, Kubernetes, CI/CD, Splunk, Git, GitLab, GitHub, Claude Code, Cursor | **AI/ML**: scikit-learn, XGBoost, OpenCV, Ollama | **Blockchain**: Solidity, Hyperledger Besu, Web3j, ethers.js, Web3.js | **Database**: PostgreSQL, MySQL, SQLite, MongoDB, NoSQL, Qdrant, pgvector, TimescaleDB, BigQuery, Oracle | **Networking/Protocols**: REST, REST APIs, WebSockets, MQTT | **Operating Systems**: Windows, Linux